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Pigazzi

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(54) **TRIGGER-TYPE DISPENSING HEAD FOR A DISPENSING DEVICE FOR PASTY PRODUCTS SUCH AS TOOTHPASTES**

(58) **Field of Classification Search**

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(Continued)

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(56)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 190 days.

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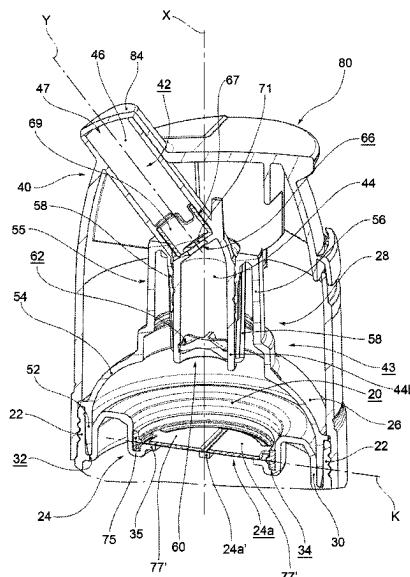
ABSTRACT

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A47K 5/12 (2006.01)
B05B 11/10 (2023.01)

(52) **U.S. Cl.**
CPC **A47K 5/1201** (2013.01); **B05B 11/1001** (2023.01); **A47K 5/1208** (2013.01);
(Continued)

A dispenser for pasty products having a fixed head that has a dispensing duct for dispensing a product, a pump with an elastically deformable cap, a manually-operated actuator, a separating wall with a suction inlet, and a suction valve cooperating with the suction inlet. The suction valve has a valve body and a plurality of tabs that bend and open the suction inlet in a product suction step, and close the suction inlet in a product dispensing step.

10 Claims, 9 Drawing Sheets



(52) **U.S. Cl.**

CPC *B05B 11/10* (2023.01); *B05B 11/1009*
(2023.01); *B05B 11/1011* (2023.01); *B05B*
11/1029 (2023.01); *B05B 11/103* (2023.01)

(58) **Field of Classification Search**

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11/1012; B05B 11/1028; B05B 11/1029;
B05B 11/1001; B65D 83/765
USPC 222/383.1, 372, 386, 387, 378; 401/152,
401/156, 158, 188 R

See application file for complete search history.

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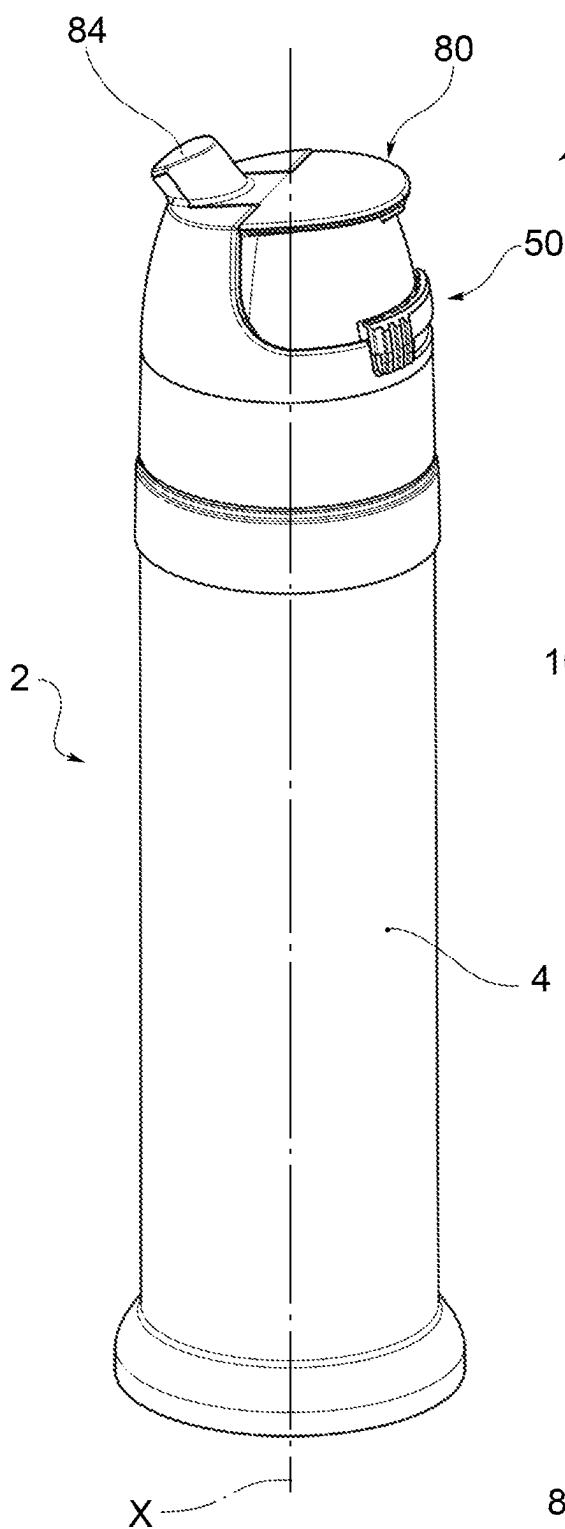


FIG. 1

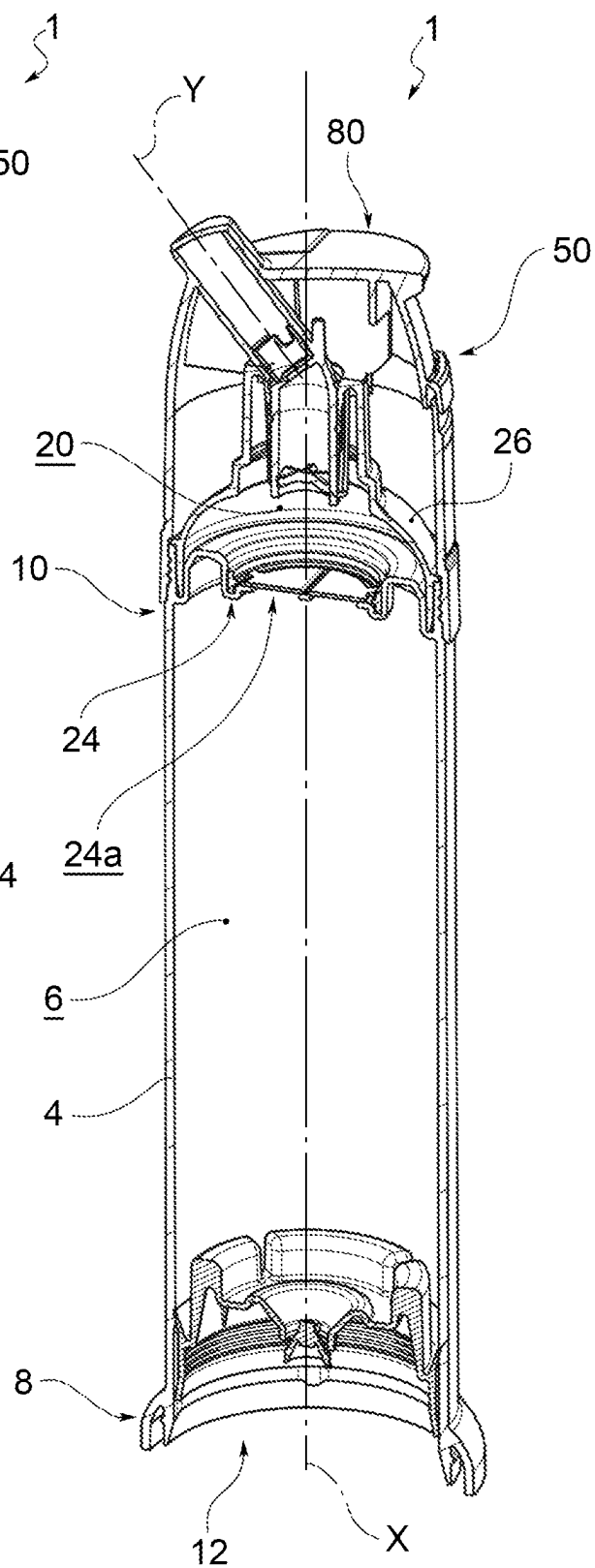


FIG. 2

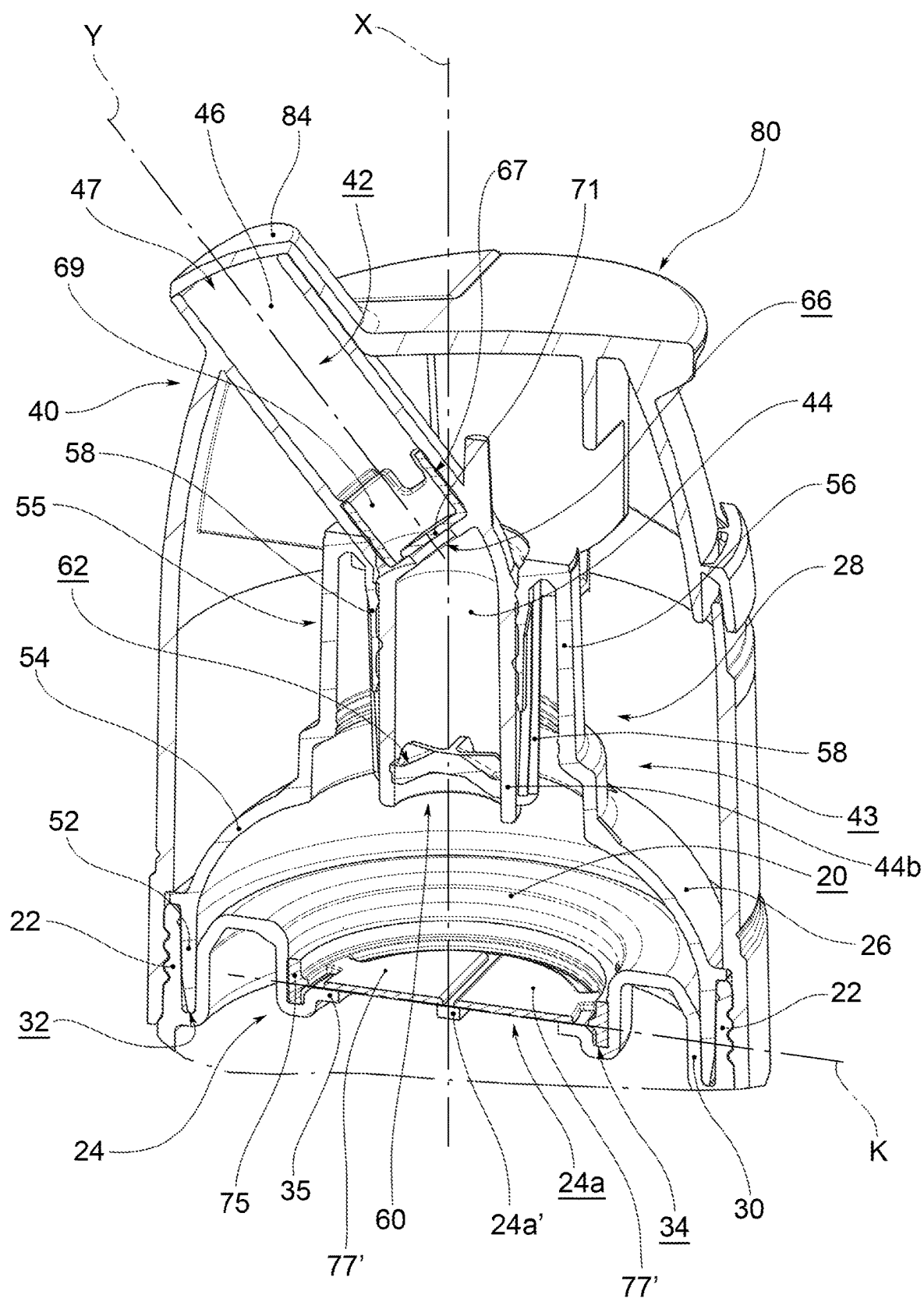


FIG.3a

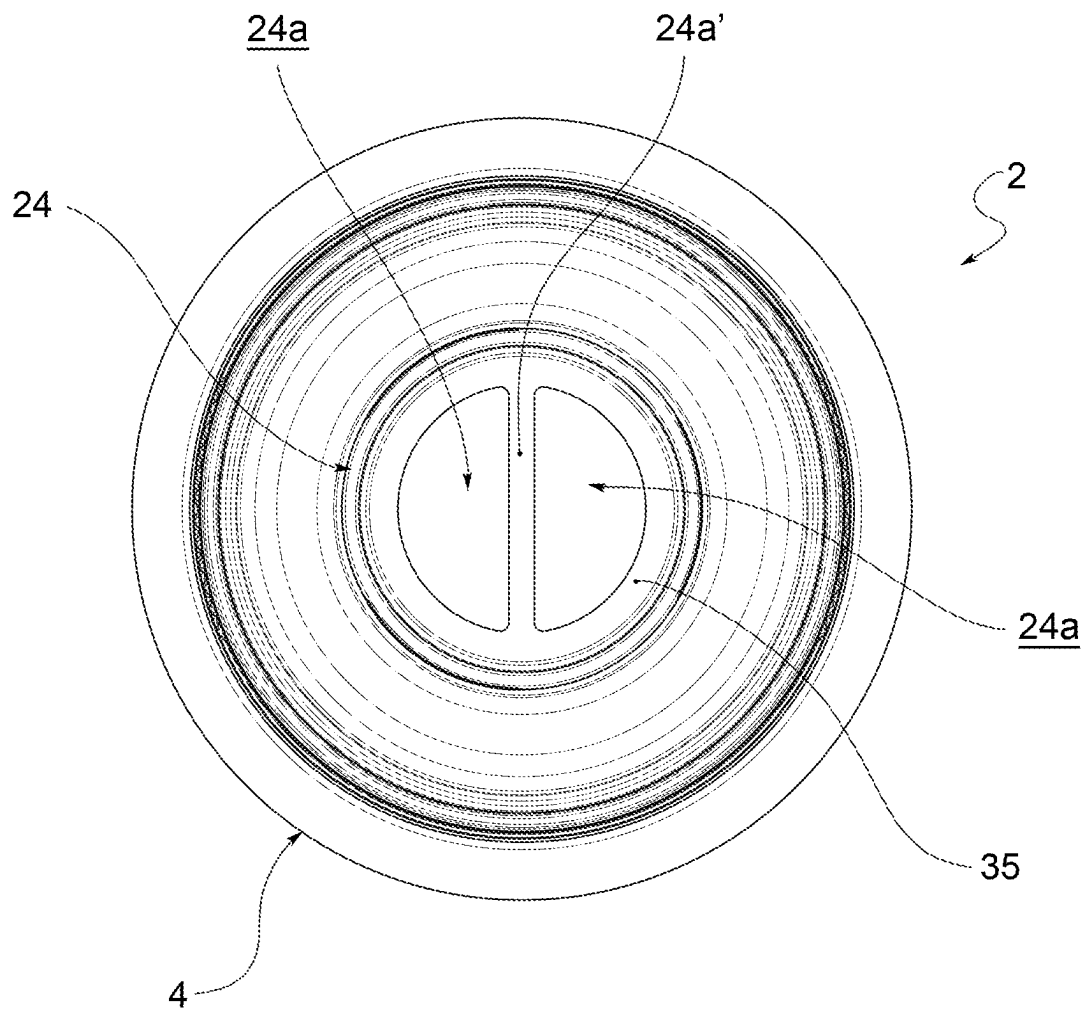


FIG.3b

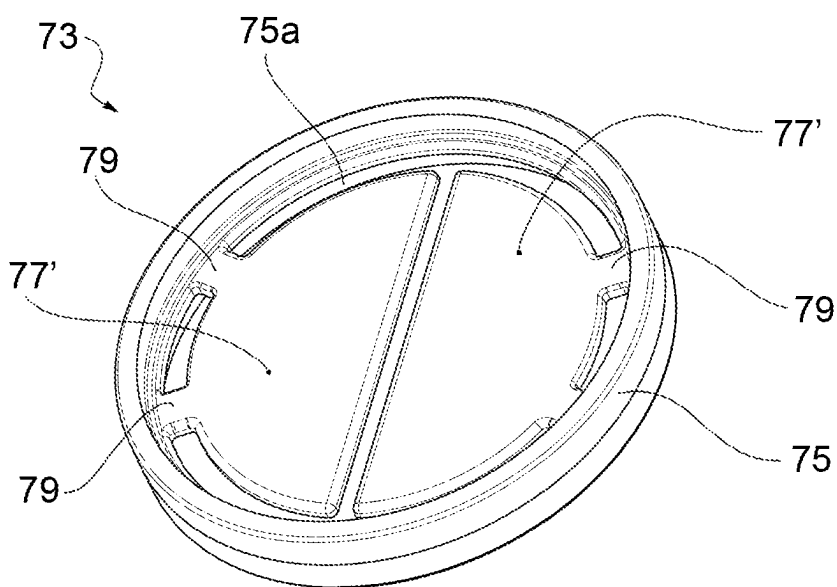


FIG. 4a

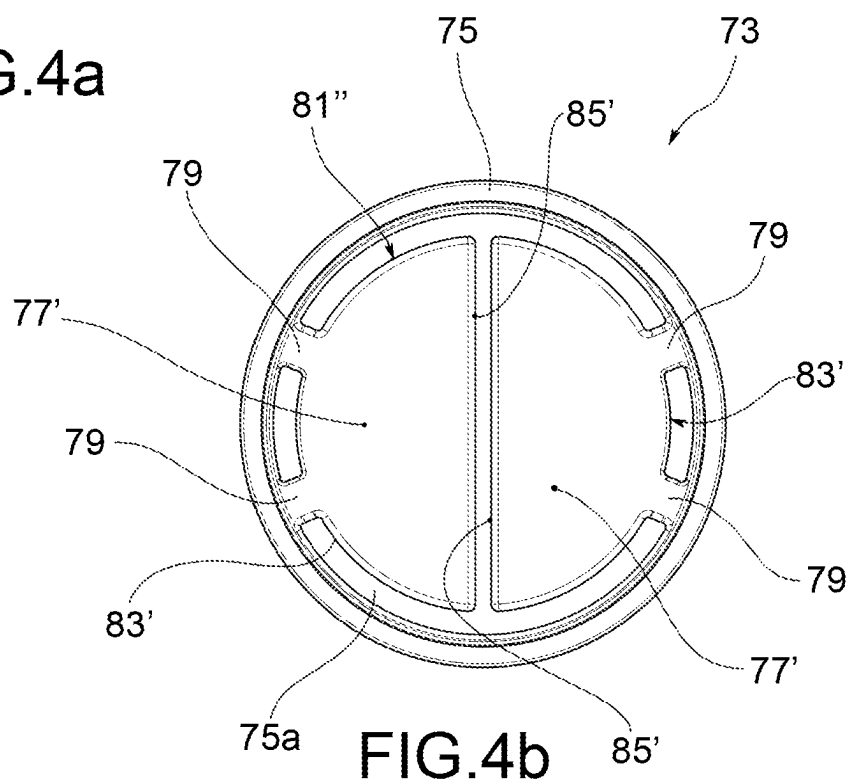


FIG. 4b

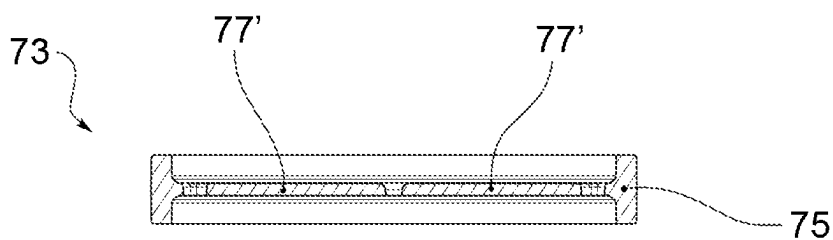


FIG. 4c

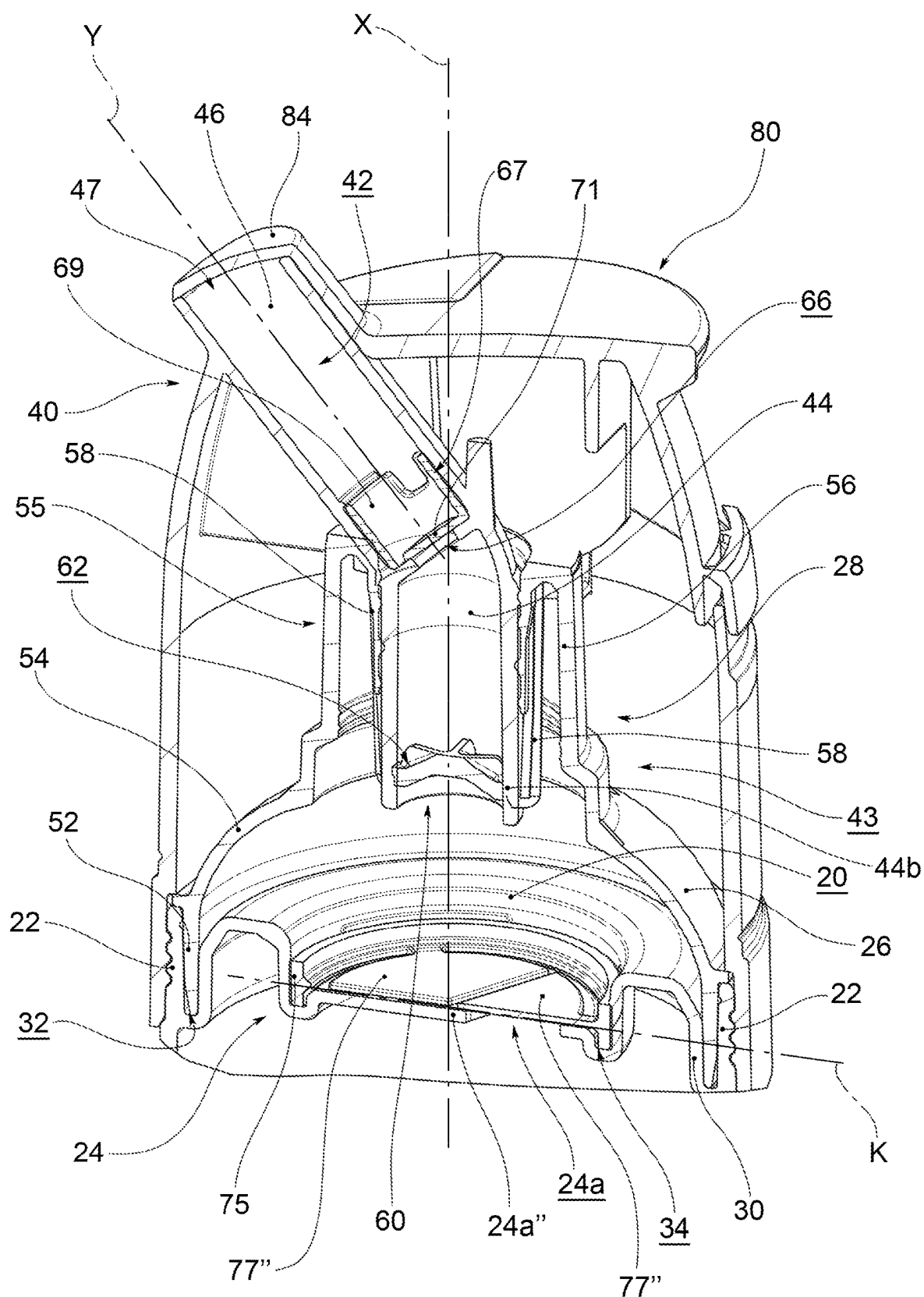


FIG.5a

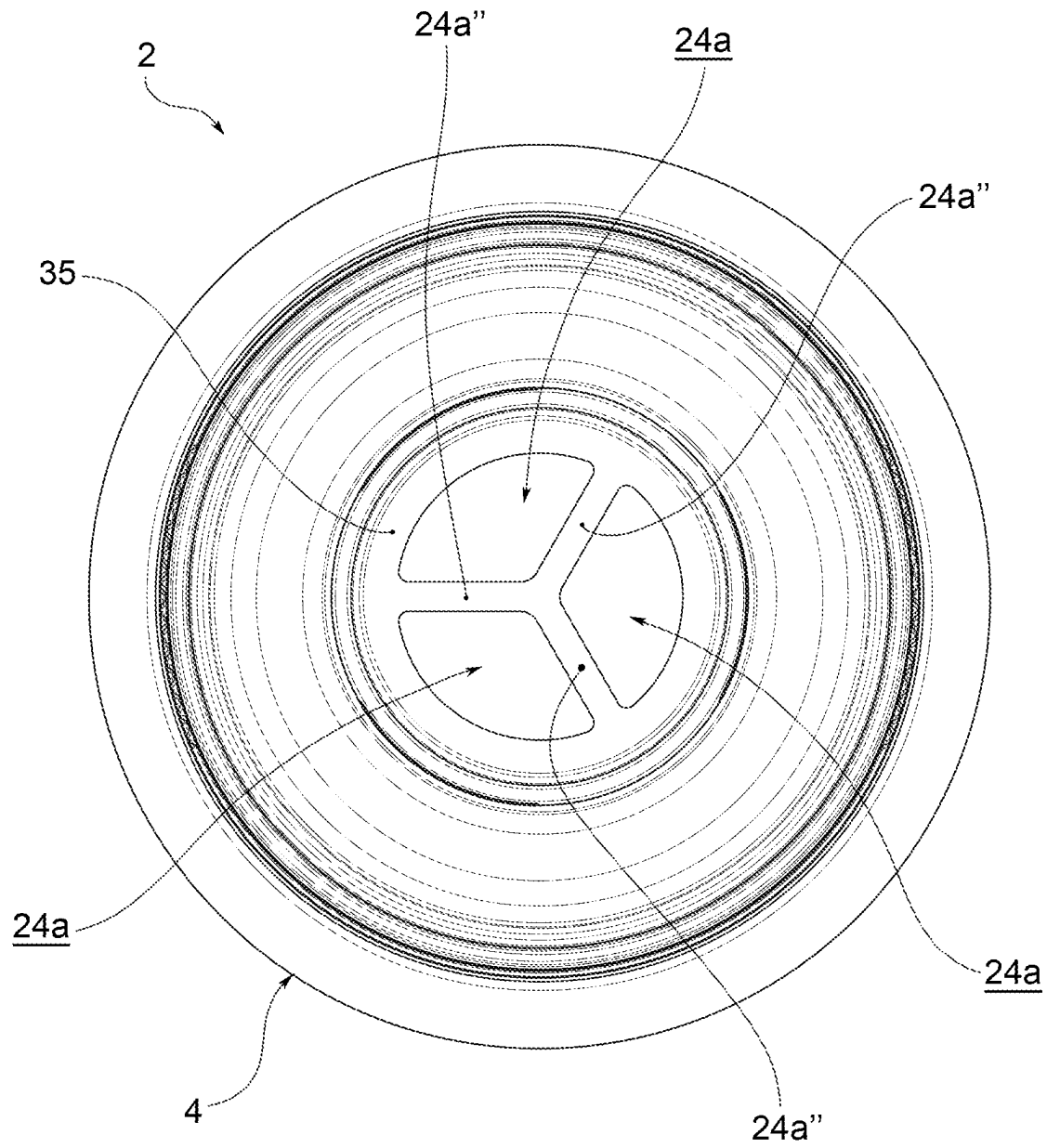


FIG.5b

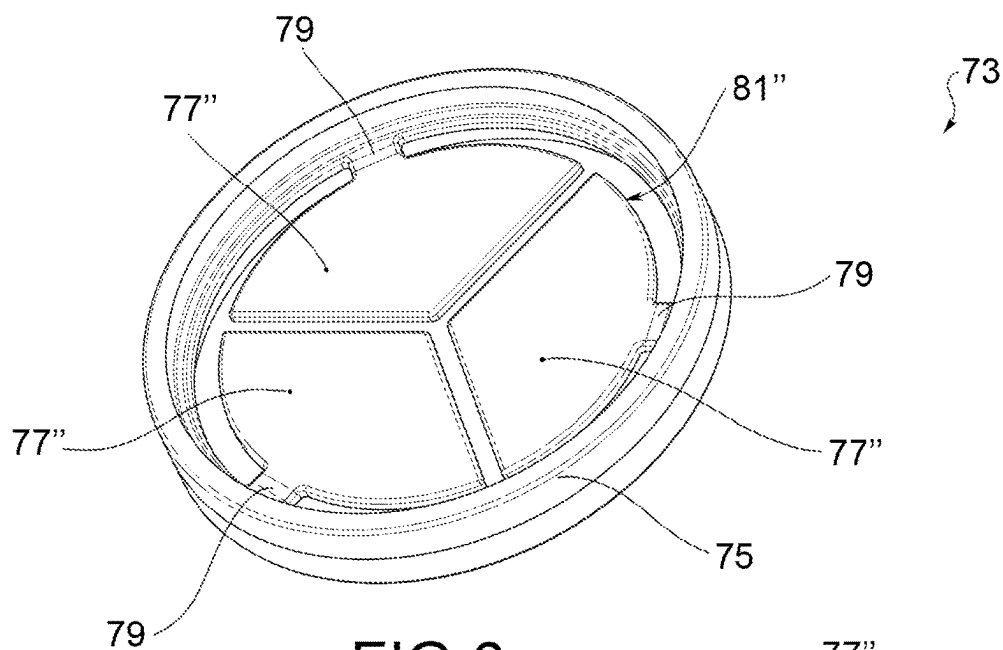


FIG. 6a

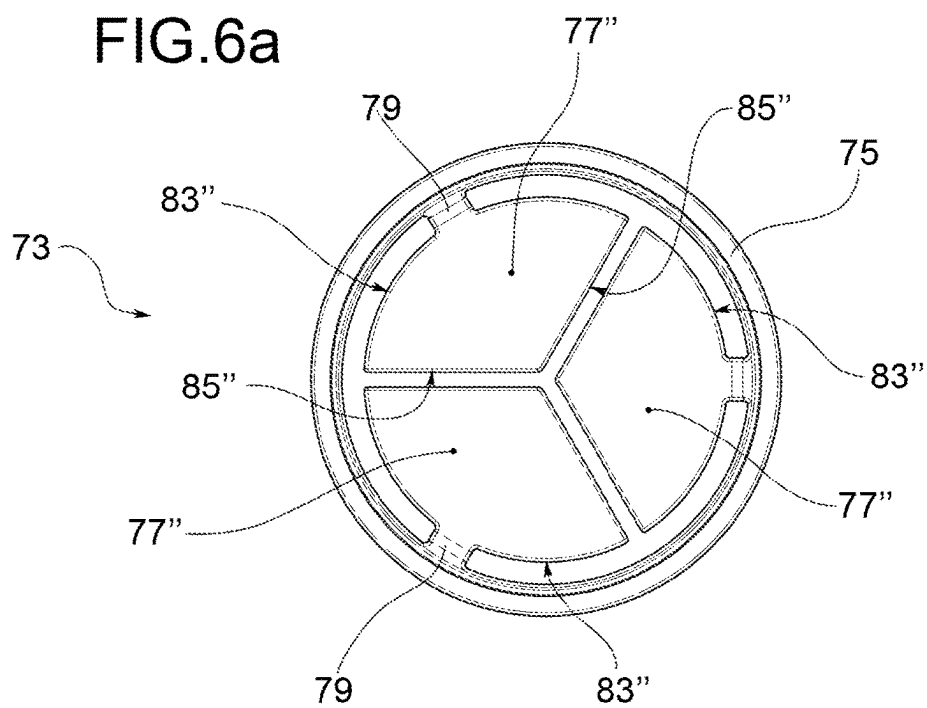


FIG. 6b

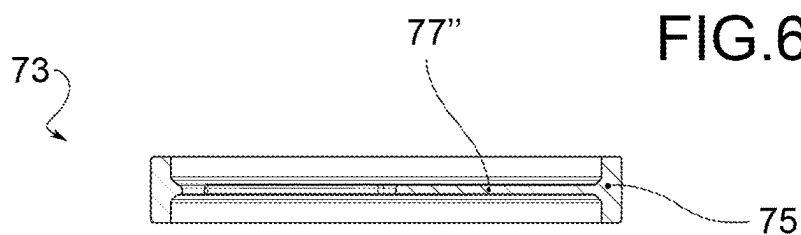


FIG. 6c

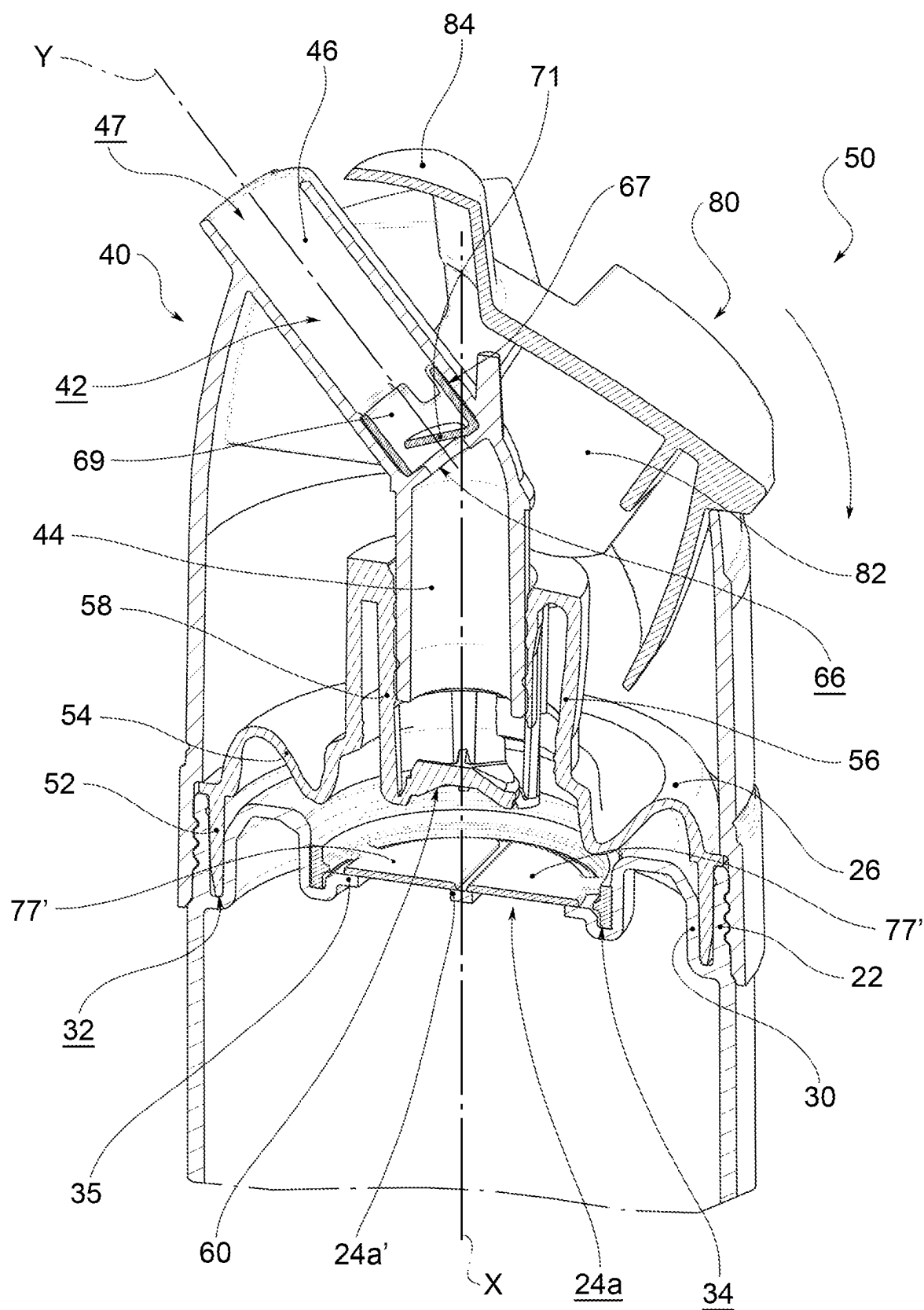


FIG. 7

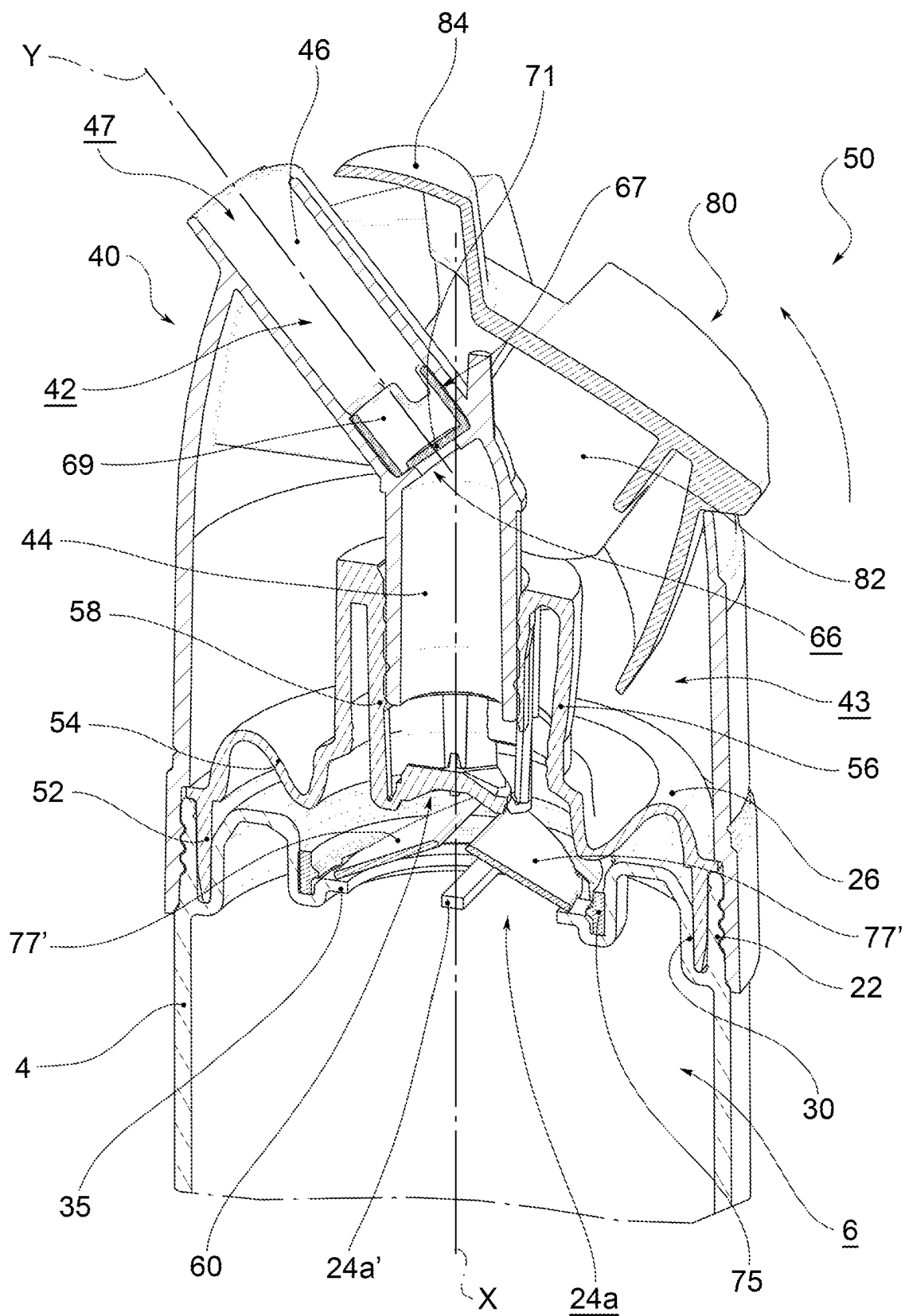


FIG.8

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TRIGGER-TYPE DISPENSING HEAD FOR A DISPENSING DEVICE FOR PASTY PRODUCTS SUCH AS TOOTHPASTES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a National Stage Application of International Patent Application No. PCT/IB2022/052085, having an International Filing Date of Mar. 9, 2022, which claims priority to Italian Application No. 102021000007847 filed Mar. 30, 2021, the entire contents of which are hereby incorporated by reference herein.

FIELD OF THE INVENTION

The subject matter of the present invention is a pushbutton dispenser for pasty products, particularly toothpastes.

BACKGROUND OF THE INVENTION

Such dispensers have been known for some time, and the Applicant itself holds the patent family coming under document U.S. Pat. No. 4,821,926. Such dispensers have been and continue to be widely known for certain advantages that set them apart from normal toothpaste tubes, such as, for example, ease of use, the ability to dispense practically all of the toothpaste without any remaining, and other advantages.

There are essentially two types of these dispensers on the market.

A first type, used especially for high-viscosity pastes, provides a metal stop ring on the bottom of the tube which, when the pushbutton is released (suction phase), advances together with the piston toward the top of the tube and prevents the piston from recoiling toward the bottom of the tube when the pushbutton is pressed (dispensing phase). An embodiment example is described in document U.S. Pat. No. 4,154,371.

A second type, used in particular for medium- or low-viscosity pastes, comprises parts having lips or diaphragms that work as check valves, and therefore the piston does not have to cooperate with a stop ring. An embodiment example is described in document U.S. Pat. No. 4,402,431.

This latter type is quickly becoming widespread because it is possible to make all the components out of plastic materials belonging to the same family. As such, these dispensers are easily recyclable and meet current environmental protection requirements. For example, the Applicant is the holder of Italian patent application Nos. 10 2020 000 025 597 and 10 2020 000 025 594.

SUMMARY OF THE INVENTION

The purpose of the present invention is to make a dispenser for pasty products, especially toothpastes, that meets the requirements of the sector, particularly concerning the use of plastic materials, and allows for effective dispensing, especially with regard to suction of the product.

This purpose is achieved by a head assembly for dispensers and by a dispenser as described and claimed herein. Advantageous embodiments of the present invention are also described.

BRIEF DESCRIPTION OF THE FIGURES

The features and advantages of the dispenser according to this invention will become apparent from the following

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description, given as a non-limiting example in accordance with the figures in the accompanying drawings, wherein:

FIG. 1 depicts a dispenser for pasty products according to an embodiment of the present invention:

FIG. 2 is a cross-sectional view of the dispenser in FIG. 1:

FIG. 3a is a cross-sectional view of a dispensing head of the dispenser according to one embodiment of the invention, in a rest configuration:

FIG. 3b is a plan view of a container tube of the dispenser according to an embodiment of the invention:

FIGS. 4a to 4c show a valve body of a suction valve of the dispenser according to one embodiment of the present invention;

FIG. 5a is a cross-sectional view of a dispensing head of the dispenser according to another embodiment of the invention, in a rest configuration:

FIG. 5b is a plan view of a container tube of the dispenser according to another embodiment of the invention:

FIGS. 6a to 6c show a valve body of a suction valve of the dispenser according to another embodiment of the present invention;

FIG. 7 is a cross-sectional view of the dispensing head in a product dispensing configuration; and

FIG. 8 is a cross-sectional view of the dispensing head in a product suction configuration.

DETAILED DESCRIPTION

In reference to the figures in the appended drawings, number 1 refers to a dispenser for pasty products, particularly toothpastes, as a whole according to an embodiment of the present invention.

The dispenser 1 comprises a container tube 2, preferably made as a single part from plastic material, for example a polymer material belonging to the polyolefin group, such as polypropylene (PP).

The container tube 2 comprises an annular tube wall 4, typically circular and cylindrical, having a central axis X. The tube wall 4 delimits a periphery of a tube compartment 6 intended to contain the pasty product. The tube wall 4 goes from a lower end 8, which is itself open and through which the tube wall 4 is filled with the pasty product, to an upper end 10 to which a dispensing head 50, to be discussed later, is applied.

The dispenser 1 further comprises a piston 12 received into the tube compartment 6 by an interference fit, designed to slide axially from the lower end 8, where it is located when the tube compartment 6 is full of product, to the upper end 10 as the product is dispensed.

Preferably, the piston 12 is made of a plastic material, preferably a polymer material belonging to the polyolefin group, for example high-density polyethylene (HDPE).

The dispensing head 50 further comprises a main chamber 20 placed at the upper end 10 of the container tube 2 for containing the product that is about to be dispensed.

According to a preferred embodiment, the main chamber 20 is delimited at the bottom by a separating wall 24 occupying the tube compartment 6 in the diameter direction, and at the top by a cap 26 of a pump 28. The separating wall 24 is provided with a suction inlet 24a capable of establishing communication between the tube compartment 6 and the main chamber 20.

The suction inlet 24a sits on an imaginary plane K perpendicular to the main axis X.

According to an embodiment of the invention, the separating wall **24** is a single part together with the container tube **2**.

For example, the separating wall **24** comprises an annular outer portion **30**, preferably coaxial with the tube wall **4**, radially close to an end section **22** of the tube wall **4** with which it forms an annular seat **32** for the pump **28**.

In addition, the separating wall **24** preferably comprises an annular functional seat **34**, preferably placed inside the outer portion **30**, to accommodate a support for a valve body, which will be discussed later herein, and a support crown **35**, placed radially inside the functional seat **34**, to support said valve body.

The dispensing head **50** further comprises a fixed head **40** which can be placed on the container tube **2** at the upper end **10**. For example, the fixed head **40** is snapped into the end section **22** of the tube wall **4**.

The fixed head **40** comprises a dispensing duct **42** for connecting the main chamber **20** to the outside environment for dispensing the product.

Preferably, the dispensing duct **42** comprises an initial section **44** that opens directly into the main chamber **20**, and an end section **46** connected to the initial section **44** and ending with an opening **47** for dispensing the product.

Preferably, the initial section **44** lies along a main axis, preferably coaxial to the tube wall **4**, i.e., coaxial to the central axis X: the end section **46**, however, is preferably at an angle and therefore lies along a dispensing axis Y that is inclined with respect to the central axis X.

Once the fixed head **40** is placed on the container tube **2**, a space **43** is delimited between said fixed head **40** and the separating wall **24** in which is placed the pump **28**, which is applied to and cooperates with the separating wall **24** and the dispensing duct **42**.

The pump **28**, consisting of a hollow shell, is made as a single part from plastic material, for example a polymer material belonging to the polyolefin group, such as low-density polyethylene (LDPE).

Preferably, the pump **28** comprises an annular connection portion **52** consisting, for example, of a preferably continuous cylindrical circular wall, designed to be housed by interference fit in the seat **32** formed by the tube wall **4** and the separating wall **24**.

The pump **28** further comprises a dome-shaped elastically deformable cap **54** diverging toward the connection portion **52**. Said cap **54** is designed to be deformed with respect to a rest condition in order to reduce the volume of the main chamber **20** and cause the product to be dispensed. Preferably, the cap **54** is connected to the connection portion **52** at a lower end.

The pump **28** further comprises, at the end opposite the end of the connection portion **52**, a pump neck **55** connected to the cap **54** and protruding axially therefrom. The pump neck **55** is slidably inserted into the initial section **44** of the dispensing duct **42**.

For example, the pump neck **55** consists of a tubular column portion **56**, preferably coaxial with the central axis X, and a tubular guide portion **58** connected to the column portion **56** and contained radially and preferably also axially therein. In other words, the guide portion **58** forms a flap with respect to the column portion **56**, thus making therewith an annular upside-down-U section.

The dispensing head **50** further comprises an intermediate valve or shutoff valve allowing product to go from the main chamber **20** to the dispensing duct **42** during a product dispensing phase, and preventing product from coming out

of the main chamber **20** and from going to the dispensing duct **42** when the container tube is being filled with the product.

For example, the pump **28** comprises a bottom **60** supported by the pump neck **55** and in particular by the guide portion **58**: the bottom **60** is provided with a plurality of bottom openings **62** arranged circumferentially. Once the pump **28** and the fixed head **40** are placed on the container tube **2**, the initial section **44** of the dispensing duct **42** is inside the guide portion **58** and, in an initial rest condition, the appropriately shaped free end **44b** of said initial section **44** closes said bottom openings **62** of the bottom **60**. The free end **44b** and the bottom **60** provided with bottom openings **62** form, for example, an intermediate or shutoff valve **64**.

The dispensing head **50** further comprises an upper or delivery valve placed downstream of the intermediate valve for allowing product to go from the main chamber **20** to the outdoor environment during a product dispensing phase, and for preventing the return of product or outdoor environment air to the main chamber **20** during a product suction phase.

For example, a delivery port **66** is placed at the connection between the initial section **44** and the end section **46** of the dispensing duct **42**, a bowl **67** housed in the end section **46** cooperating therewith, said bowl **67** consisting of an annular cup wall **69** and a flexible cup flap **71** for closing the delivery port **66**, acting as a shutter. The delivery port **66** and the cup flap **71** constitute an example of said upper valve or delivery valve.

The dispensing head **50** further comprises a lower or suction valve for allowing product to go from the tube compartment **6** to the main chamber **20** during a product suction phase, and for preventing the return of product from the main chamber **20** to the tube compartment **6** during a product dispensing phase.

To that end, according to the invention, the dispensing head **50** comprises a valve body **73** that cooperates with the suction inlet **24a**.

Preferably, the valve body **73** comprises a support **75** consisting, for example, of an annular wall, received at least partially into the functional seat **34** of the separating wall **24**, and a plurality of tabs **77'**, **77''** contained in the support **75** which, by cooperating with each other and with the separating wall **24**, are capable of closing off the suction inlet **24a**.

In addition, the valve body **73** preferably comprises a plurality of hinge portions **79**, wherein each hinge portion **79**, which is elastically deformable, connects a tab **77** to the support **75** and, by deforming, allows said tab to move from a closed configuration of the suction inlet **24a** during the product suction phase to an open configuration of the suction inlet **24a** during the product dispensing phase.

In particular, by means of said hinge portions **79**, each tab **77'**, **77''** is arranged in a cantilevered fashion inside the support **75**, that is, spaced from said support so as to form an annular gap **75a** therewith, except at said hinge portions **79**.

According to a first embodiment variant (FIGS. **3a**, **3b**, and from **4a** to **4c**) in which the suction inlet **24a** is divided by a crosspiece **24a'**, preferably along the diameter, the valve body **73** comprises two tabs **77'**: for example, each tab **77'** is connected to the support **75** by two hinge portions **79**.

For example, each tab **77'** has the shape of a half moon, provided with a perimeter edge **81''** having an arched section **83'**, preferably in the shape of a semicircle, facing toward the support **75**, and a straight section **85'** facing the straight section of the other tab.

In the closed configuration, the straight sections **85'** of the tabs **77** press against the crosspiece **24a'** of the suction inlet **24a**, whereas the gap **75a** between the arched sections **83'** and the support **75** rests on the crown **35** of the separating wall **24**. In the open configuration, however, the tabs **77'** bend due to the deformation of the hinge portions **79** and/or the tab itself, thus opening the suction inlet **24a**.

According to another embodiment variant (FIGS. **5a**, **5b**, and from **6a** to **6c**) in which the suction inlet **24a** is divided by crosspieces **24a''**, preferably along a radius, the valve body **73** comprises three tabs **77''**: for example, each tab **77''** is connected to the support **75** by a single hinge portion **79**.

For example, each tab has a triangular shape, such as a lobed shape, provided with a perimeter edge **81''** having an arched section **83''**, preferably in the shape of an arc of a circle, facing toward the support **75**, and two straight sections **85''** facing the straight sections of the other tabs.

In the closed configuration, the straight sections **85''** of the tabs **77''** press against the crosspieces **24a''** of the suction inlet **24a**, whereas the gap **75a** between the arched sections **83''** and the support **75** rests on the crown **35** of the separating wall **24**.

In the open configuration, however, the tabs **77''** bend due to the deformation of the hinge portions **79** and/or the tab itself, thus opening the suction inlet **24a**.

The dispenser head **50** further comprises an actuator **80**, for example in the form of a button, held so as to be movable such as by rotation or translation from the fixed head **40**, and designed to influence the pump **28** so as to cause deformation of the cap **26** and product dispensing. For example, the actuator **80** comprises an influence portion **82** in contact with the connection area between the column portion **56** and the guide portion **58**, such that the squeezing thereof causes deformation of the cap **26**.

Preferably, the actuator **80** also comprises a closing portion **84** for closing the opening **47** of the dispensing duct **42** in a rest configuration of the dispenser.

The actuator **80** is made as a single part from plastic material, for example a polymer material chosen from the polyolefin group, such as polypropylene (PP).

In an initial or rest configuration (FIGS. **3a** and **5a**), the pump **28** is in an initial non-deformed condition, the cup flap **71** closes off the delivery port **66** (delivery valve closed), the tabs **77', 77''** are in contact with the crosspieces **24a', 24a''** and with the support crown **35** (suction valve closed), and the bottom **60** is engaged with the lower end of the initial section **44** in such a way that the bottom openings **62** are blocked (shutoff valve closed). In addition, the closing portion **84** of the actuator **80** blocks the opening **47** of the dispensing duct **42**.

A dispensing phase (FIG. **7**) takes place by actuation of the actuator **80** by gradually deforming the pump **28** and in particular the cap **54**.

When the actuator **80** is actuated, the closing portion **84** is disengaged at least partially from the opening **47**, leaving it free to dispense the product.

During the dispensing phase, the cap **54** of the pump **28** is deformed, guided along the main axis X along which the guide portion **58** travels. An attendant reduction of the volume of the main chamber **20** occurs: the overpressure caused by the squeezed product keeps the suction valve **76** in the closed configuration. In other words, the tabs **77', 77''** are push tabs with sealed contact with the crosspieces **24a', 24a''** and with the support crown **35**, thus closing the suction inlet **24a**. This prevents the product contained in the main chamber **20** from returning to the tube compartment **6** of the tube **2**.

At the same time, due to the thrust of the actuator **80**, the column portion **56** lowers, preferably guided in this movement by the guide portion **58**, which slides as it is guided in contact with the dispensing duct **42** and in particular with the initial section **44** thereof.

The lowering of the column portion **56** causes the bottom **60** to lower and disengage from the initial section **44** of the dispensing duct **42**: in particular, the lower end of the initial section **44** comes out through the bottom openings **62**. The shutoff valve **64** thus changes to the open configuration and the product contained in the main chamber **20** comes through the bottom openings **62** in the bottom **60**.

At the same time, due to the action of the product which is now located between the bottom **60** and the initial section **44** of the dispensing duct **42**, the cup flap **71** moves away from the delivery port **66**. The delivery valve thus changes to the open configuration and the product may come out through the opening **47** of the dispensing duct **42**.

When the actuator **80** reaches the end of travel, the dispensing phase ends and, after the actuator **80** is released, a suction phase occurs (FIG. **8**), during which time a predefined quantity of product contained in the tube compartment **6** enters the main chamber **20**.

In particular, when the actuator **80** is released, the pump **28** tends to return to the non-deformed condition and influences the actuator in that direction, which tends to return to the initial condition. Consequently, the volume of the main chamber **20** tends to increase and the resulting partial vacuum causes the tabs **77', 77''** to bend, disengaging from the crosspieces **24a', 24a''**. A predefined quantity of product contained in the tube compartment **6** is thus drawn into the main chamber **20**.

At the same time, the cup flap **71** recloses the delivery port **66** such that the delivery valve returns to the closed configuration. This ensures that the product in the dispensing duct **42** between the delivery valve and the opening **47** does not return to the main chamber **20**, nullifying the suction of new product from the tube compartment **6**.

At the same time, the closing portion **84** of the actuator **80** tends to return to the initial condition in which the opening **47** is blocked, thereby cutting off the flow of dispensed product and facilitating separation of the dispensed quantity.

Once the cap **54** has returned to the non-deformed condition, the bottom **60** also returns to the initial position in which it engages with the initial section **44**; the shutoff valve **64** thus returns to the closed configuration and, since the overpressure in the main chamber has ceased, the suction valve **76** also returns to the closed configuration.

According to an embodiment variant of the invention, the separating wall is a part made separately from the tube **4** and subsequently applied thereto.

Innovatively, the dispenser according to the present invention is particularly well suited for dispensing pasty products and in particular has an especially effective suction phase thanks to the special arrangement of the suction inlet and the configuration of the suction valve.

It is clear that a person skilled in the art, in order to satisfy current needs, may make modifications to the dispenser described above, said modifications all being contained within the scope of protection as defined in the following claims.

What is claimed is:

1. A manually-operable head assembly for a dispenser for pasty products, comprising:

a fixed head adapted to be applied to a container tube of the dispenser, said fixed head comprising a dispensing

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duct provided with an opening for dispensing a product and with an initial section extending along a main axis;
 a pump comprising an elastically deformable cap guided under deformation along said main axis;
 an actuator movably supported by the fixed head and adapted to be manually operated to deform at least partially the pump and achieve dispensing of the product;
 a separating wall cooperating with the pump, adapted to delimit at least partially a main chamber with the elastically deformable cap and to separate said main chamber from a tube compartment of the dispenser, wherein said separating wall comprises a suction inlet adapted to put the tube compartment in communication with the main chamber, lying on an imaginary plane orthogonal to the main axis; and
 a suction valve adapted to allow the product to pass from the tube compartment to the main chamber during a product suction step and to prevent the product from returning from the main chamber to the tube compartment during a product dispensing step;
 wherein said suction valve comprises a valve body that cooperates with the suction inlet and comprises a plurality of tabs which, in the product suction step, flex and open the suction inlet and, in the product dispensing step, close the suction inlet,
 wherein the valve body further comprises an annular support, supported by the separating wall, and a plurality of hinge portions, wherein each hinge portion of said plurality of hinge portions, being elastically deformable, joins at least one tab of the plurality of tabs to the annular support, and
 wherein by said hinge portions, each tab of the plurality of tabs is arranged in a cantilever manner inside the annular

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support, spaced apart from said annular support to form an annular gap therewith, except at said hinge portions.
 2. The manually-operable head assembly of claim 1, wherein the separating wall comprises a support crown, and wherein said annular gap is arranged on said support crown.
 3. The manually-operable head assembly of claim 1, wherein the separating wall comprises at least one cross-piece that divides the suction inlet.
 4. The manually-operable head assembly of claim 3, wherein, during the product dispensing step, the tabs cooperate with said at least one crosspiece to close the suction inlet.
 5. The manually-operable head assembly of claim 3, wherein, during the product suction step, the tabs separate from said at least one crosspiece to open the suction inlet.
 6. The manually-operable head assembly of claim 3, wherein the suction inlet is divided by a single crosspiece and the valve body comprises two tabs.
 7. The manually-operable head assembly of claim 3, wherein the suction inlet is divided by three crosspieces and the valve body comprises three tabs.
 8. A manually-operable dispenser for pasty products, comprising:
 a container tube to contain a product; and
 a manually-operable head assembly according to claim 1, applied to the container tube.
 9. The manually-operable dispenser of claim 8, wherein the separating wall is in a single piece with the container tube.
 10. The manually-operable dispenser of claim 8, wherein the separating wall is a separate component from the container tube and applicable thereto.

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